

(a) failure kernel: the value of the first-matching declared rule (rule key A–I, below)

		Recon.	Res. dev.	Init. access	Execution	Persistence	Priv. esc.	Stealth	Def. imp.	Cred. access	Discovery	Lat. move.	C2	Collection	Exfiltration	Impact
preparation	Reconnaissance		0.7 _G	0.4 _B	0.05 _C	0.05 _C	0.05 _C	0.05 _C	0.05 _C	0.05 _C	0.05 _C	0.05 _C	0.05 _C	0.05 _C	0.05 _C	0.05 _C
	Resource development	0.7 _G		0.3 _I	0.3 _I	0.3 _I	0.3 _I	0.3 _I	0.3 _I	0.3 _I	0.3 _I	0.3 _I	0.3 _I	0.3 _I	0.3 _I	0.3 _I
intrusion	Initial access	0.9 _F	0.9 _F		0.02 _A	0.02 _A	0.02 _A	0.02 _A	0.02 _A	0.02 _A	0.02 _A	0.02 _A	0.02 _A	0.02 _A	0.02 _A	0.02 _A
	Execution	0.25 _D	0.25 _D	0.7 _G		0.35 _H	0.35 _H	0.35 _H	0.35 _H	0.35 _H	0.35 _H	0.35 _H	0.35 _H	0.35 _H	0.35 _H	0.35 _H
post-intrusion	Persistence	0.25 _D	0.25 _D	0.25 _D	0.35 _E		0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.35 _H	0.35 _H	0.35 _H
	Privilege escalation	0.25 _D	0.25 _D	0.25 _D	0.35 _E	0.7 _G		0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.35 _H	0.35 _H	0.35 _H
	Stealth	0.25 _D	0.25 _D	0.25 _D	0.35 _E	0.7 _G	0.7 _G		0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.35 _H	0.35 _H	0.35 _H
	Defense impairment	0.25 _D	0.25 _D	0.25 _D	0.35 _E	0.7 _G	0.7 _G	0.7 _G		0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.35 _H	0.35 _H	0.35 _H
	Credential access	0.25 _D	0.25 _D	0.25 _D	0.35 _E	0.7 _G	0.7 _G	0.7 _G	0.7 _G		0.7 _G	0.7 _G	0.7 _G	0.35 _H	0.35 _H	0.35 _H
	Discovery	0.25 _D	0.25 _D	0.25 _D	0.35 _E	0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.7 _G		0.7 _G	0.7 _G	0.35 _H	0.35 _H	0.35 _H
	Lateral movement	0.25 _D	0.25 _D	0.25 _D	0.35 _E	0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.7 _G		0.7 _G	0.35 _H	0.35 _H	0.35 _H
	Command and control	0.25 _D	0.25 _D	0.25 _D	0.35 _E	0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.7 _G	0.7 _G		0.35 _H	0.35 _H	0.35 _H
actions on objectives	Collection	0.25 _D	0.25 _D	0.25 _D	0.35 _E	0.9 _F	0.9 _F	0.9 _F	0.9 _F	0.9 _F	0.9 _F	0.9 _F	0.9 _F		0.7 _G	0.7 _G
	Exfiltration	0.25 _D	0.25 _D	0.25 _D	0.35 _E	0.9 _F	0.9 _F	0.9 _F	0.9 _F	0.9 _F	0.9 _F	0.9 _F	0.9 _F	0.7 _G		0.7 _G
	Impact	0.25 _D	0.25 _D	0.25 _D	0.35 _E	0.9 _F	0.9 _F	0.9 _F	0.9 _F	0.9 _F	0.9 _F	0.9 _F	0.9 _F	0.7 _G	0.7 _G	

(b) lifecycle-distance kernel $d(a, b)$ over the four consensus stages, the signed stage offset $\Delta = s(b) - s(a)$ as subscript: $d = 1$ for $|\Delta| \leq 1$, $\gamma\Delta^{-1}$ forward, $\delta|\Delta|^{-1}$ backward, $d < z$ reads as 0 ($\gamma = 0.25$, $\delta = 0.25$, $z = 0.1$)

		Recon.	Res. dev.	Init. access	Execution	Persistence	Priv. esc.	Stealth	Def. imp.	Cred. access	Discovery	Lat. move.	C2	Collection	Exfiltration	Impact
preparation	Reconnaissance		1 ₀	1 ₊₁	1 ₊₁	0.25 ₊₂	0.25 ₊₂	0.25 ₊₂	0.25 ₊₂	0.25 ₊₂	0.25 ₊₂	0.25 ₊₂	0.25 ₊₂	0 ₊₃	0 ₊₃	0 ₊₃
	Resource development	1 ₀		1 ₊₁	1 ₊₁	0.25 ₊₂	0.25 ₊₂	0.25 ₊₂	0.25 ₊₂	0.25 ₊₂	0.25 ₊₂	0.25 ₊₂	0.25 ₊₂	0 ₊₃	0 ₊₃	0 ₊₃
intrusion	Initial access	1 ₋₁	1 ₋₁		1 ₀	1 ₊₁	1 ₊₁	1 ₊₁	1 ₊₁	1 ₊₁	1 ₊₁	1 ₊₁	1 ₊₁	0.25 ₊₂	0.25 ₊₂	0.25 ₊₂
	Execution	1 ₋₁	1 ₋₁	1 ₀		1 ₊₁	1 ₊₁	1 ₊₁	1 ₊₁	1 ₊₁	1 ₊₁	1 ₊₁	1 ₊₁	0.25 ₊₂	0.25 ₊₂	0.25 ₊₂
post-intrusion	Persistence	0.25 ₋₂	0.25 ₋₂	1 ₋₁	1 ₋₁		1 ₀	1 ₀	1 ₀	1 ₀	1 ₀	1 ₀	1 ₀	1 ₊₁	1 ₊₁	1 ₊₁
	Privilege escalation	0.25 ₋₂	0.25 ₋₂	1 ₋₁	1 ₋₁	1 ₀		1 ₀	1 ₀	1 ₀	1 ₀	1 ₀	1 ₀	1 ₊₁	1 ₊₁	1 ₊₁
	Stealth	0.25 ₋₂	0.25 ₋₂	1 ₋₁	1 ₋₁	1 ₀	1 ₀		1 ₀	1 ₀	1 ₀	1 ₀	1 ₀	1 ₊₁	1 ₊₁	1 ₊₁
	Defense impairment	0.25 ₋₂	0.25 ₋₂	1 ₋₁	1 ₋₁	1 ₀	1 ₀	1 ₀		1 ₀	1 ₀	1 ₀	1 ₀	1 ₊₁	1 ₊₁	1 ₊₁
	Credential access	0.25 ₋₂	0.25 ₋₂	1 ₋₁	1 ₋₁	1 ₀	1 ₀	1 ₀	1 ₀		1 ₀	1 ₀	1 ₀	1 ₊₁	1 ₊₁	1 ₊₁
	Discovery	0.25 ₋₂	0.25 ₋₂	1 ₋₁	1 ₋₁	1 ₀	1 ₀	1 ₀	1 ₀	1 ₀		1 ₀	1 ₀	1 ₊₁	1 ₊₁	1 ₊₁
	Lateral movement	0.25 ₋₂	0.25 ₋₂	1 ₋₁	1 ₋₁	1 ₀	1 ₀	1 ₀	1 ₀	1 ₀	1 ₀		1 ₀	1 ₊₁	1 ₊₁	1 ₊₁
	Command and control	0.25 ₋₂	0.25 ₋₂	1 ₋₁	1 ₋₁	1 ₀	1 ₀	1 ₀	1 ₀	1 ₀	1 ₀	1 ₀		1 ₊₁	1 ₊₁	1 ₊₁
actions on objectives	Collection	0 ₋₃	0 ₋₃	0.25 ₋₂	0.25 ₋₂	1 ₋₁	1 ₋₁	1 ₋₁	1 ₋₁	1 ₋₁	1 ₋₁	1 ₋₁	1 ₋₁		1 ₀	1 ₀
	Exfiltration	0 ₋₃	0 ₋₃	0.25 ₋₂	0.25 ₋₂	1 ₋₁	1 ₋₁	1 ₋₁	1 ₋₁	1 ₋₁	1 ₋₁	1 ₋₁	1 ₋₁	1 ₀		1 ₀
	Impact	0 ₋₃	0 ₋₃	0.25 ₋₂	0.25 ₋₂	1 ₋₁	1 ₋₁	1 ₋₁	1 ₋₁	1 ₋₁	1 ₋₁	1 ₋₁	1 ₋₁	1 ₀	1 ₀	

(c) the failure weight set = (a) × (b), as committed (v4_failure_only)

		Recon.	Res. dev.	Init. access	Execution	Persistence	Priv. esc.	Stealth	Def. imp.	Cred. access	Discovery	Lat. move.	C2	Collection	Exfiltration	Impact
preparation	Reconnaissance		0.7	0.4	0.05	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0	0	0
	Resource development	0.7		0.3	0.3	0.075	0.075	0.075	0.075	0.075	0.075	0.075	0.075	0	0	0
intrusion	Initial access	0.9	0.9		0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.005	0.005	0.005
	Execution	0.25	0.25	0.7		0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.0875	0.0875	0.0875
post-intrusion	Persistence	0.0625	0.0625	0.25	0.35		0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.35	0.35	0.35
	Privilege escalation	0.0625	0.0625	0.25	0.35	0.7		0.7	0.7	0.7	0.7	0.7	0.7	0.35	0.35	0.35
	Stealth	0.0625	0.0625	0.25	0.35	0.7	0.7		0.7	0.7	0.7	0.7	0.7	0.35	0.35	0.35
	Defense impairment	0.0625	0.0625	0.25	0.35	0.7	0.7	0.7		0.7	0.7	0.7	0.7	0.35	0.35	0.35
	Credential access	0.0625	0.0625	0.25	0.35	0.7	0.7	0.7	0.7		0.7	0.7	0.7	0.35	0.35	0.35
	Discovery	0.0625	0.0625	0.25	0.35	0.7	0.7	0.7	0.7	0.7		0.7	0.7	0.35	0.35	0.35
	Lateral movement	0.0625	0.0625	0.25	0.35	0.7	0.7	0.7	0.7	0.7	0.7		0.7	0.35	0.35	0.35
	Command and control	0.0625	0.0625	0.25	0.35	0.7	0.7	0.7	0.7	0.7	0.7	0.7		0.35	0.35	0.35
actions on objectives	Collection	0	0	0.0625	0.0875	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9		0.7	0.7
	Exfiltration	0	0	0.0625	0.0875	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.7		0.7
	Impact	0	0	0.0625	0.0875	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.7	0.7	

rule key (match order; value × d gives the cell in (c)):

A ia_gate_foothold = 0.02
B recon_gate_initial_access = 0.4
C recon_gate_deep = 0.05
D preintrusion_damper = 0.25
E execution_damper = 0.35

F backward = 0.9
G lateral = 0.7
H forward_from_foothold = 0.35
I forward_from_prep = 0.3